PREDICTING LINEAR RELATIONAL DECODABILITY WITHOUT THE RELATION

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ABSTRACT

Linear relational embeddings (LREs) approximate a language model's subject-to-object mapping with an affine function. Their faithfulness varies enormously across relations – near zero for PERSON_FATHER, at ceiling for COUNTRY_CAPITAL_CITY. We ask what determines that variation and show it is substantially predictable from a readout that never sees the relation. We estimate a novel form of a Jacobian lens, the *differential Jacobian lens*: instead of model outputs for "France", we instead contrastively predict outputs for "France" that are not also outputs for "Spain". After estimating this differential Jacobian lens, we measure how highly it ranks the model's answer for a relation – without using any relation-specific information in estimation and prediction. Across 28 relations of the LRE dataset on Olmo-3-7B this relation-agnostic score correlates with LRE faithfulness at $\rho = 0.66$ ($p < .01$). This suggests that linear decodability may be largely a property of what a subject's representation already carries, rather than of the relation being decoded. After attempting to move the unit of analysis from the relation to the individual fact, we conclude that new relation datasets and analysis methods are needed to address several problems in existing datasets, most prominently semantically weak first tokens (e.g., most predicted universities' first token is "University") and subject ambiguity (e.g., the LLM associates obscure actor names with non-actor entities).

Configuration note. The main-text numerical analyses retain the historical protocol. Appendix A gives the current verified comparison across seven models using the shared split and newline-prefixed neutral prompt, and supersedes the historical baseline comparisons.

1 INTRODUCTION

Hernandez et al. (2024) showed that a language model's mapping from a subject to the object of a relation is often well approximated by an affine function of the subject's hidden state – a *linear relational embedding* (LRE). They also showed that this works far better for some relations than others. On Olmo-3-7B, we reproduce the same spread: an LRE fitted on COUNTRY_CAPITAL_CITY agrees with the model's own prediction 75% of the time while one fitted on PERSON_FATHER agrees 9% of the time.

Why? The natural reading is that some relations are "more linear" than others – that linearity is a property of the relation. We report evidence that it is better understood as a property of the *subject's representation*. Concretely, we show that one can predict how faithful an LRE will be *before fitting it*, and without using the relation at all.

Building on recent work by Gurnee et al. (2026), we introduce the *differential Jacobian lens*, which evaluates the Jacobian at individual reference entities and applies it to *differences* between entity representations: *France* minus *Spain* rather than *France* alone. What the subtraction leaves is what is specific to this entity among others of its kind, and the Jacobian carries that residual forward to the vocabulary. The construction changes what the readout is about: from a category to an individual. The question it answers – *what is this model ready to say about this entity, as against others of its kind?* – is not specific to relations.

We estimate the differential Jacobian lens under a single neutral prompt shared by every relation, at one layer fixed in advance, and ask how highly it ranks the model's own answer. The lens sees

no relation template, no objects and no (subject, object) pairs, yet its score correlates with LRE faithfulness across 28 relations at $\rho = 0.66$.

We make four contributions. (i) We introduce the differential Jacobian lens, a supervision-free readout of what a model is disposed to say about an individual entity. (ii) We turn it into a relation-agnostic diagnostic of linear decodability and show that it predicts LRE faithfulness. (iii) We show the prediction holds up under leave-one-domain-out and under controls for object cardinality and answer-token frequency. (iv) After attempting to move the unit of analysis from the relation to the individual fact, we provide an analysis of several problems in existing datasets – most prominently semantically weak first tokens and subject ambiguity – that hinder further progress on understanding LREs in language models.

2 Background

An LRE for relation r approximates the model's computation as

$$o \;\approx\; \beta\, W s + b_\beta, \tag{1}$$

where s is the hidden state at the subject's final token and o the output state. Write f for the model's own map between the two: f sends the subject's hidden state at an intermediate layer ℓ to the final-layer state at the last position. We average local first-order approximations of f over a small set $\{s_i\}$ of reference subjects and scale their subject-dependent terms by β:

$$W = \mathrm{mean}_i\, J_i, \qquad b_\beta = \mathrm{mean}_i[\, f(s_i) - \beta J_i s_i \,], \tag{2}$$

where J_i is the Jacobian of f at s_i. At $\beta = 1$, this is the averaged first-order approximation (Hernandez et al., 2024).

The scalar β corrects a systematic weakness of the first-order estimate. A Jacobian taken at a reference subject describes the model's map only in a neighborhood of that subject; extrapolated to a different subject it points in roughly the right direction but is too short. This matters because the prediction decomposes into a subject-dependent term and a constant: expanding b_β gives

$$\beta W s + b_\beta \;=\; \mathrm{mean}_i[\, f(s_i) + \beta J_i (s - s_i) \,], \tag{3}$$

in which only the second term depends on which subject is being decoded. Left unscaled, the shared anchor $\mathrm{mean}_i\, f(s_i)$ dominates and the operator returns much the same object for every subject. Setting $\beta > 1$ amplifies the subject-specific contribution relative to that anchor, and so does change which object is ranked first. Hernandez et al. (2024) fit β per relation alongside the layer; we follow their protocol and select it from $\{0.5, 1, 1.5, 2, 3, 5\}$.

Faithfulness is the rate at which decoding $\beta W s + b_\beta$ over the full vocabulary reproduces the *model's own* top-1 prediction – not the gold object. It is therefore a measure of how well an affine map captures what the model does. Faithfulness defined this way is the quantity we predict.

3 A relation-agnostic Jacobian lens

For an entity e (e.g., France) and a set of reference entities $\{i\}$ of the same type (e.g., Spain, Italy), we take the Jacobian–vector product at each reference and average:

$$v_e \;=\; \frac{1}{n} \sum_i J_i\, (s_e - s_i). \tag{4}$$

Here J_i is the Jacobian of f at reference i, the same map as in Eq. 2. As discussed above, subtracting a same-type reference cancels it, leaving the entity-specific residual, and the Jacobian then carries that residual forward to the vocabulary. Our differential Jacobian lens therefore reads what this entity, as opposed to others of its kind, makes the model disposed to say.

We never form J_i explicitly, but obtain each Jacobian–vector product $J_i u$ by the standard double-backward construction (Pearlmutter, 1994), at the cost of a constant number of forward and backward passes; this is what makes Eq. 4 affordable at $d_{\text{model}} = 4096$ (Appendix B). Each J_i is evaluated under a *neutral* prompt, `"{} is known for"`, identical for every relation.

We decode v_e over the full vocabulary and record the rank of m_e, the model's own top-1 answer for relation r. Writing $\mathrm{rk}_{v_e}(m_e)$ for that rank, the diagnostic for a relation is

$$A_r \;=\; -\,\mathrm{median}_e\, \log_{10} \mathrm{rk}_{v_e}(m_e), \tag{5}$$

so higher A_r means the lens agrees with the model more.

Three properties make A_r a non-trivial predictor of faithfulness. The prompt is neutral and shared, so no relation template enters. The differences in Eq. 4 are subject-minus-subject – *France* minus *Spain*, never *France* minus *Paris* – so no relation-specific (subject, object) pair is used in estimation. The layer is not relation-specific either. It is fixed at 17 for all relations (Appendix C).

Two prompts. Our experiments use two different prompts, for two different purposes. The *neutral prompt*, `"{} is known for"`, names no relation and is the same for all 28 relations; it is the only prompt the lens ever sees, and it is where the hidden states in Eq. 4 are taken. The *in-context prompt* is relation-specific: eight (subject, object) examples of the relation in question, followed by the query subject without its object, so that the model's next token is its answer (§4). It is used for the LRE and for the model's own prediction, never for the lens.

Relation-specificity therefore enters in exactly one place: the *target*. What the lens is scored against is the model's top-1 under the in-context prompt – the same quantity faithfulness is defined against – so the two are commensurable, while everything on the lens side of the comparison remains relation-agnostic.

4 Experimental setup

Model. Olmo-3-7B, in `float32` with eager attention, as the Jacobian–vector products require both (Appendix B).

Relations. The LRE dataset (Hernandez et al., 2024) contains 47 relations. Our prompt presupposes an entity subject, so we fixed a subject-type filter in advance using the dataset's declared `domain_name` field: the 28 relations with named-entity subjects (*person*, *country*, *name*, *company*, *landmark*, *superhero*, *city*, *pokemon*, *product*, *star*). Excluded are relations over words as words (*verb*, *adjective*, *word*) and over abstracts (*occupation*, *task*, *characteristic*, *degree*), for which `"give is known for"` is a category error. No other relation was excluded.

Measurement. Both quantities are read off the same prompt, so that they are comparable. Following Hernandez et al. (2024), a relation is presented to the model *in context*: eight (subject, object) examples of that relation are concatenated, one per line, and the query subject is appended without its object, so that the model's next token is its answer. This is what defines "the model's own prediction" for both measures. The subjects of a relation are shuffled once and partitioned into three disjoint sets: eight in-context examples, eight reference subjects from which W and b_β are estimated, and the remainder as scored test subjects.

Faithfulness is then recomputed for all 28 relations under this protocol. Layer and β are chosen on a 6×6 grid – six candidate layers for the subject state $(5, 11, 14, 17, 20, 23)$ crossed with six values of β $(0.5, 1, 1.5, 2, 3, 5)$ – by the mean over the test subjects. Hernandez et al. (2024) likewise fit both per relation; the six layers span the network at roughly even spacing and the β range brackets the values they report.

The lens scores 11–20 subjects per relation, depending on how many remain once the in-context and reference sets are removed. For the results in §5, the lens and the LRE sample their splits independently, so the particular subjects entering each set differ between the two measurements. Independent splits introduce sampling variation into this comparison. Because the LRE hyperparameters are selected on the scored subjects, its reported faithfulness is an in-sample estimate; these results do not establish performance on a separate held-out evaluation set.

Crucially, the in-context prompt is used only to obtain the *target*; the lens itself is always computed under the neutral prompt of §3.

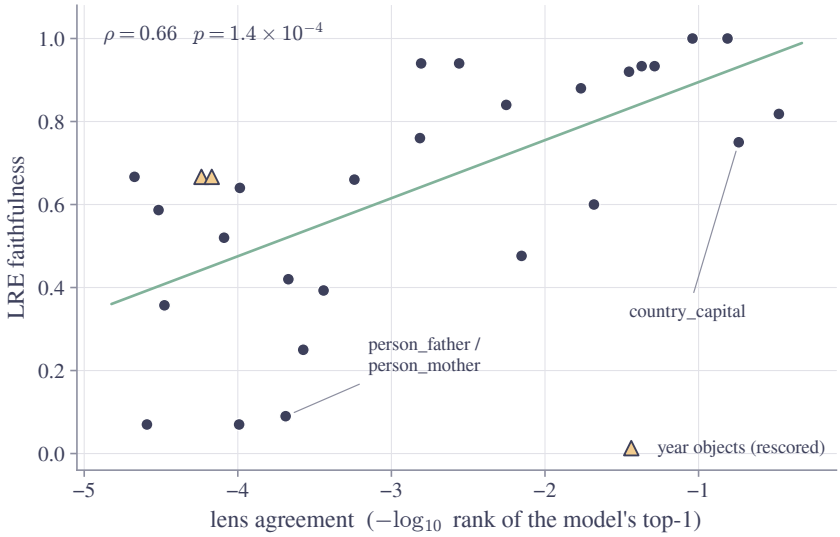


Figure 1: LRE faithfulness against the relation-agnostic lens score, 28 relations. The lens sees no relation template, no objects, and no (subject, object) pairs, and uses one fixed layer throughout. Triangles mark the two relations with numeric objects, rescored as described in Appendix E.

5 Results

Figure 1 shows the result. The relation-agnostic score predicts LRE faithfulness at Spearman $\rho = 0.658$ ($p = 1.4 \times 10^{-4}$); a permutation test agrees (Appendix D).

The correlation is not carried by cardinality. Relations differ greatly in how many distinct objects they have; k, the mean number of subjects per object, ranges from 1.0 to 189. High k makes faithfulness easier, and indeed k predicts faithfulness on its own ($\rho = 0.43$, $p = 0.02$). But k is essentially uncorrelated with the lens score ($\rho = -0.03$). Removing k *raises* the correlation to $\rho = 0.744$ ($p = 10^{-5}$): k and the lens are independent predictors of faithfulness, so removing the variance attributable to k sharpens rather than weakens the lens signal.[1]

Nor by answer frequency. A relation whose answers are frequent tokens might be easy for both measures. We estimate the model's prior over each answer token by averaging its log-probability across ten neutral contexts. This predicts neither faithfulness ($\rho = -0.03$) nor the lens score ($\rho = -0.02$), and removing it leaves the correlation unchanged at 0.658. The correlation is 0.746 with both cardinality and answer frequency removed.

Nor by one domain. Eleven of the 28 relations have *person* subjects, raising the worry that one domain (i.e., one subject type) drives the result. Holding out each subject domain in turn leaves $\rho \in [0.570, 0.700]$, median 0.656.

Unchanged findings under an alternative scoring protocol. We reran the analysis in Figure 1 under an alternative protocol: a single shared split and objects restricted to those their first token identifies. A relation meets the restriction if, after discarding every object whose first token it shares with another object of the same relation, enough subjects remain to form a split: eight in-context examples, eight reference subjects and at least one test subject. Three relations fall below that

[1] All variables here are ordinal, so we work throughout with ranks. To remove the effects of a set of covariates, we rank-transform the lens score, faithfulness, and each covariate; regress the ranked lens score and ranked faithfulness separately on the ranked covariates using ordinary least squares with an intercept; and report the Pearson correlation between the resulting residual vectors. This is the standard partial Spearman correlation, and it asks how much of the lens–faithfulness association remains once everything the covariates can linearly explain about either variable has been removed.

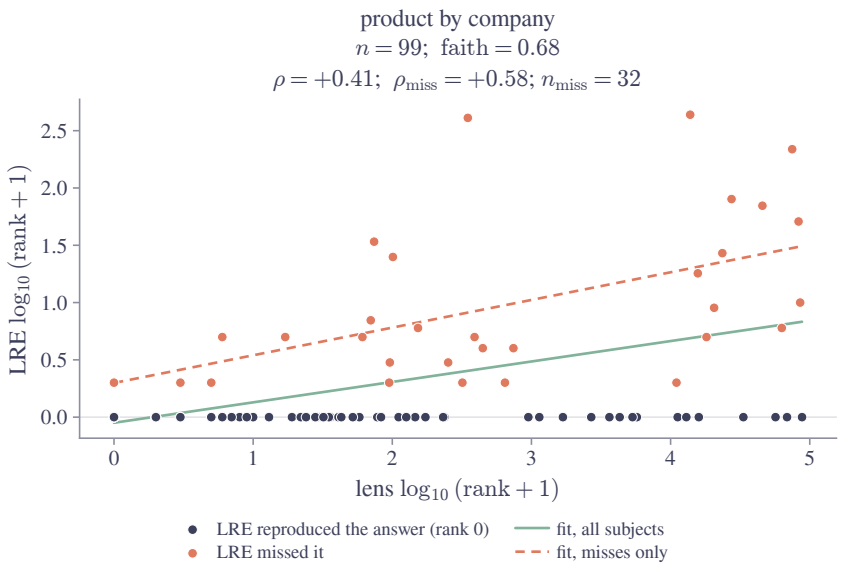


Figure 2: Rank-rank analysis of individual facts for PRODUCT_BY_COMPANY. x-axis: lens $\log_{10}(\text{rank} + 1)$. y-axis: LRE $\log_{10}(\text{rank} + 1)$; ranks are zero-based. Faithfulness: 0.68. Green solid regression line over all points (n=99, $\rho = .41$). Orange dashed regression line over the orange points, the subjects the LRE missed (n=32, $\rho = .58$).

threshold and are dropped – the two with numeric objects, whose years collapse onto shared leading digits, and PERSON_BAND_LEAD_SINGER. Over the remaining 25 the alternative protocol gives $\rho = 0.644$ ($p = 0.0005$), against 0.658 over 28 here. We return to first-token collapse, and to why restricting the gold objects is only a partial remedy, in §7.

6 WHERE THE ACCOUNT DEGRADES

What the lens reads is what an entity is *salient* for: a relation is linearly decodable to the extent that its answer is already part of what the subject's representation makes the model disposed to say. The relations that deviate from the trend are therefore exceptions to the account. Here, we ask whether these relations are unstructured.

Salience, measured independently of the lens. Our account would be circular if salience were read off the lens, since the lens is itself a salience readout. We therefore measure it by the model's own free generation. For each test subject we greedily generate 90 tokens from each of `"{} is best known for"` and `"Here are some facts about {}. First,"`, and record whether any distinctive word of the gold object appears in either continuation.[2] This is a forward pass under different prompts, and involves no Jacobian.

We refer to this rate as the object's *prominence*. Across the 14 relations for which it can be measured it ranges from 1.00 (CITY_IN_COUNTRY) and 0.50 (COUNTRY_CAPITAL_CITY) down to 0.13 (PERSON_FATHER), 0.05 (SUPERHERO_ARCHNEMESIS) and 0.03 (PERSON_MOTHER). It predicts how often the lens surfaces the gold object in its top five: Spearman $\rho = 0.614$ ($p = 0.019$, $n = 14$). Table 1 gives both quantities per relation. The salience interpretation is thus supported by a measurement that is independent of the quantity it explains.

One relation is an outlier: STAR_CONSTELLATION has prominence 0.60 yet the lens surfaces the constellation for only 5% of its stars. We suspected an artifact, since a Bayer designation contains the

[2]Titles, particles and words shared with the subject's own name are excluded, so that *The* Joker is not matched by any occurrence of "the" and *Johann Georg Kant* is not matched by a passage about Immanuel Kant. The lens's own prompt cannot be used here: `"{} is known for"` makes this model emit a multiple-choice quiz rather than prose, so the object is never named.

Table 1: Object prominence, measured by the model's own free generation and therefore independently of the lens, against how often the lens surfaces the gold object in its top five. Spearman $\rho = 0.614$ ($p = 0.019$, $n = 14$). *: star_constellation, the outlier discussed in §6.

relation	prom.	n	gold∈lens	n
city_in_country	1.00	5	0.65	17
person_plays_pro_sport	0.60	40	0.35	20
star_constellation*	0.60	40	0.05	20
country_capital_city	0.50	8	0.56	16
landmark_in_country	0.47	40	0.25	20
product_by_company	0.33	40	0.05	20
person_occupation	0.30	40	0.10	20
country_currency	0.29	14	0.00	20
person_university	0.28	29	0.00	20
country_largest_city	0.25	8	0.25	16
person_father	0.13	39	0.06	18
company_hq	0.12	40	0.00	20
superhero_archnemesis	0.05	39	0.05	19
person_mother	0.03	40	0.00	20

constellation in the Latin genitive (*Eta Persei* for *Perseus*), which our word filter does not remove; but prominence on the 19 subjects without such overlap is 0.63, slightly higher rather than lower, so this explanation fails. What the lens returns for these subjects is category vocabulary – *constellation*, *brightness*, *stellar* – rather than the particular constellation. Excluding the relation would raise the correlation to 0.725.

We now return to our question whether relations that deviate from the overall trend (the lens does not predict linearity well) are unstructured. They are not. Order the relations by their residual from the least-squares fit of faithfulness on the lens score in Figure 1; a residual is in units of faithfulness, so $+0.30$ means thirty points more faithful than the lens score predicts. Both tails are then organized by the size of the answer space. The largest positive residuals – faithful despite a low lens score – are LANDMARK_ON_CONTINENT ($+0.30$, $k = 189$), NAME_GENDER ($+0.29$, $k = 9.5$) and PERSON_NATIVE_LANGUAGE ($+0.26$, $k = 30.6$). In each, the LRE chooses among a handful of possible answers, and can be faithful without the answer being salient for any particular subject: no first name is chiefly "known for" its gender, but there are only two answers to pick between. The largest negative residuals – unfaithful despite the lens – are PERSON_FATHER (-0.43), PERSON_MOTHER (-0.41) and COMPANY_CEO (-0.32), all with $k \approx 1$ and answer spaces in the hundreds, where faithfulness is near its floor whatever the representation carries.

The account thus holds in the middle of the cardinality range and degrades predictably at both ends, in opposite directions. This is the same variable whose removal raises the correlation to 0.744 (§5), and it gives that number an interpretation: k is the principal source of deviation in the salience account rather than merely a nuisance correlate.

7 FACT-LEVEL ANALYSIS

Linearity emerges as a phenomenon at the relation level, not the fact-level – it would not make sense to investigate linearity for the facts of a relation that is not represented linearly. But once there is evidence for at least partial linearity of a relation, we can ask for individual facts how well the differential Jacobian lens predicts the degree to which they are integrated into the linear structure.

We use the setup shown in Figure 2 to investigate fact-level linearity. The figure shows that lens rank is a moderately good predictor of LRE rank for PRODUCT_BY_COMPANY ($\rho = .41$), especially if we only consider misses ($\rho = .58$). Points with an LRE rank of 0 (i.e., a hit, not a miss) are "censored": They give us no resolution because they fail to distinguish cases where the LRE predicts the correct object perfectly vs those where it barely avoids a miss. So misses may be a better diagnostic for assessing the size of the correlation.

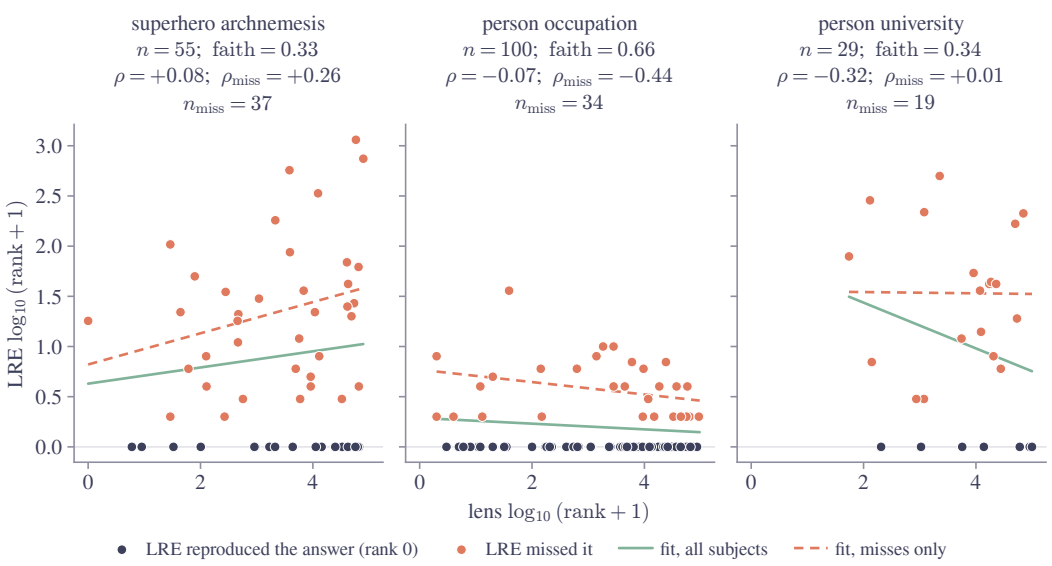


Figure 3: Rank-rank analysis for the three relations with the worst overall correlation between lens rank and LRE rank. Green solid regression lines over all points, orange dashed regression lines over the orange points, the subjects the LRE missed. All three panels share both scales.

Our original plan was to develop a fine-grained account of what differential Jacobian lens readouts predict about the linearity of individual facts. However, we found that several shortcomings of the experimental setup currently used for investigating relations in LMs make this hard. In this section, we analyze these shortcomings and give guidance for future relation datasets that may better support research on how LMs represent facts.

We conduct our analysis on the three relations that have the worst overall correlation in the rank-rank analysis: their correlations are close to zero or even negative. See Figure 3.

7.1 Semantically weak first tokens

PERSON_UNIVERSITY exemplifies the problem of *semantically weak first tokens*. Many universities have names of the form "University of". So the LM learns to predict "University" as the target token for many persons. The standard setup is to count such a prediction as correct. But this is not an effective evaluation: The correct generation of "University" is not a reliable indicator of the factual competence of the LM.

Measuring the collapse rate – the fraction of a relation's distinct objects that share a first token with another – **6 of 40 relations exceed 25%**, led by PERSON_MOTHER (61%), PERSON_FATHER (58%) and PERSON_UNIVERSITY (45%, where 29 of 69 objects begin with UNIVERSITY).

Restricting each relation to objects whose first token identifies them repairs this on the gold side. It is only a partial remedy, because faithfulness never uses gold: it compares the LRE against the *model's* top-1, so what inflates it is the model's own answer being an easy token. In PERSON_UNIVERSITY, after filtering, the model still answers UNIVERSITY for 7 of 29 subjects and the LRE agrees on every one. A complete fix would condition on the prediction being distinctive.

The direction of the bias differs by relation. Where the collapse is on the gold side it makes faithfulness easier – 41 distinct fathers all counting as JOHN – so our PERSON_FATHER figure is an upper bound, and the corrected value is lower still ($0.07 \rightarrow 0.05$). Where the model itself emits the shared token, the collapse manufactures agreement that is not retrieval at all.

We flag all of this because the failure is silent: it produces a perfect or inflated score rather than an obvious error, and it will affect any first-token evaluation whose answers share prefixes – numeric, titled (*Doctor Doom*, *The Joker*) or generic (*University of X*).

Both LRE and the lens work on the token level, the residual stream vector at the last layer, which is focused on producing the correct next token and may not represent reliable information about the tokens to follow. This single-token regime is highly problematic for the standard setup because we are effectively investigating a relation in which all objects have been shortened to their first token. This might be a nonsensical relation even to humans and it is clearly different from the intended relations whose objects are full entity names.

7.2 SUBJECT AMBIGUITY

PERSON_OCCUPATION exemplifies the problem of *subject ambiguity*. An example is Rolf Thommessen who is listed as an actor in LRE. The highest ranked token given by the lens is "whistlebl". This indicates that the most prominent entity associated with the last name Thommessen by Olmo-3-7B is Advokatfirmaet Thommessen, a law firm known for their digital whistleblowing tool. One might hope that the first name "Rolf" disambiguates which subject is intended, but apparently this is not the case. In general, the last few tokens seem to dominate which entity is represented and used by the LM. For the actor Barry Sobel the lens top-ranks "kosher", "Jewish", "Rabbi", correct associations for the Brazilian Rabbi Henry Sobel. For the actress Vondell Darr, the lens top-ranks "criminal", "murders", "felon", plausible associations for the wrongfully convicted Acacia Lyn-Darr. For the actor Simon Gosejohann, the lens top-ranks "Johann", "Hann", "German", suggesting the LM simply uses the string "Johann" as the representation. While we did not conduct an extensive analysis, it is likely that subject ambiguity is a problem for many relations whose subjects are persons. Many of those persons share the properties that result in Barry Sobel, Vondell Darr and Simon Gosejohann being misrepresented by the model.

7.3 TOWARDS BETTER FACT DATASETS

Our analysis motivates two requirements for future evaluations. First, answer scoring should distinguish complete objects rather than credit a shared, semantically weak first token. Filtering for identifying first tokens provides one diagnostic, but excludes some relations and does not ensure that the model's own prediction is distinctive. Evaluation methods that score complete answers are therefore needed for relations with unavoidable prefix overlap.

Second, a subject name should refer to the intended entity for the model under study. Dataset construction alone may not ensure this, because different models can resolve the same name differently. Evaluations should examine the entity that a model associates with each subject, document ambiguity, and distinguish errors of subject identification from errors in relational decoding. These are methodological directions motivated by the observed failures, rather than solutions evaluated in this study.

8 DISCUSSION

1 <<< hs: this section is a collection of content pieces that need to be integrated into a fluent narrative >>>

The instrument we use to show this is new, and we offer it as a contribution in its own right. A *Jacobian lens* propagates a first-order approximation of the model's own computation from an intermediate layer to the output vocabulary; recent work uses one to characterize what a model is internally disposed to say, averaging a Jacobian over a corpus and applying it to raw activations (Gurnee et al., 2026). That is the right construction for describing a model's dispositions in general. It is the wrong one for describing a single entity, because a raw activation is dominated by what every entity of its type shares – the type itself, the syntactic role, the prompt – so the entity is buried in its own category.

Why differences. The quantity we want is not what the model says about France, but what is *specific* to France among countries. A raw activation s_e is dominated by what all entities of its type share – the type itself, the syntactic role, the prompt – and a lens applied to it returns mostly that shared content.

We note that this differencing falls directly out of the rewriting of the Jacobian lens in Eq. 3, i.e., out of expressing the LRE as a constant plus a sum of $J_i(s - s_i)$ terms. Eq. 4 is that same quantity with the subject-independent anchor dropped and $\beta = 1$. This new way of looking at the Jacobian lens allows us to estimate the lens without any relation and ask whether it predicts an LRE that was fitted *with* one. This also distinguishes our lens from that of Gurnee et al. (2026), which averages a Jacobian over a corpus and applies it to raw activations.

9 LIMITATIONS

The study covers one model and 28 relations. The relationship is correlational: we show that a relation-agnostic readout predicts LRE faithfulness, not that the representation it reads is what the LRE exploits.

The account has a scope condition we can state but not yet explain (§6). It is weakest where the answer space is very small, since an LRE can then be faithful by elimination rather than by reading anything from the subject, and where the answer space is very large and each answer near-unique, since faithfulness is then close to its floor regardless. We show that these are the two tails of a single variable, but we do not have a principled account of where the boundaries lie, and with 28 relations we cannot fit one.

We also probe salience through a single neutral prompt, `"{} is known for"`. That phrasing asks what an entity is *noted* for, which need not coincide with everything its representation carries; attributes such as gender may be well encoded without being something a name is "known for". A different neutral prompt, or an average over several, might track salience more broadly.

Seven of the 28 relations have fewer than ten test items after the in-context and reference subjects are removed, so their faithfulness estimates are coarse (for three of them, quantized to thirds); a rotating-split protocol would sharpen these.

Finally, the diagnostic is defined only for relations whose subjects are entities, which excludes 19 of the dataset's 47 relations.

10 CONCLUSION

Whether a relation is linearly decodable from a language model is substantially predictable without the relation. A Jacobian lens estimated from subject differences under a neutral prompt, at a fixed layer, ranks the model's own answer highly for exactly those relations whose LREs turn out to be faithful.

The reading we favor is that salience is the mechanism rather than a confound: a relation admits a faithful linear operator to the extent that its answer is already part of what the subject's representation makes the model disposed to say. On that view the wide spread of LRE faithfulness across relations is less a fact about the relations than about the entities they are asked about – and the exceptions are informative, falling at the two ends of the answer-space range where faithfulness can be achieved, or prevented, without the subject's representation mattering much either way.

AI USE STATEMENT

OpenAI Codex assisted with conversion to the ICLR 2027 template, LaTeX formatting, editorial revisions, and automated document checks. It also assisted with auditing experimental configurations, rerunning matched baseline measurements, generating the shared-split appendix and its tables, and checking numerical provenance and document builds.

ETHICS STATEMENT

This work studies model behavior using existing relation datasets. Model predictions and the associations reported in the subject-ambiguity analysis should not be interpreted as verified attributes of the named individuals. First-token collisions and mistaken entity identification illustrate risks of using such readouts to make factual claims about people. The experiments are intended to diagnose these limitations, not to support individual profiling.

REPRODUCIBILITY STATEMENT

Section 4 describes the model, relation selection, prompts, data partitions, and hyperparameter grid. Equations 2–4 define the estimators; Appendices B, C, D, and E document the Jacobian–vector products, layer choice, permutation test, and numeric-token correction. Table 5 reports the relation-level results.

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A SHARED-SPLIT BASELINE COMPARISONS

We compare five readouts across seven models, then examine corpus-frequency baselines for OLMo 3 7B. These first two tables report Spearman correlations with LRE faithfulness across 25 relations, giving each relation equal weight. The measurements use the current shared split and supersede the historical unprefixed-prompt comparisons.

Matched protocol. After filtering objects whose first answer token does not identify a unique complete object, seed 0 selects eight demonstrations, eight references, and up to 100 test rows. All readouts receive the same test pool within a model; tokenizer-dependent filtering can change membership across models. The neutral prompt is exactly `\n{subject} is known for`. Models use pinned revisions, float32, eager attention, and one fixed layer per model. The relation prompts, model answer tokens, and selected LRE settings are held fixed.

Table 2: **Readout baselines across seven models.** Spearman correlation ρ with LRE faithfulness on 25 shared-split relations per model. All neural readouts use the same newline-prefixed neutral prompt.

Model	Differential Jacobian	Raw Jacobian	Logit lens	Differential logit lens	Attribute rate
OLMo 3 7B	0.683	**0.689**	0.234	0.326	−0.114
GPT-J 6B	**0.573**	0.567	0.531	0.394	0.002
Mistral 7B	**0.606**	0.586	0.402	0.249	−0.304
Gemma 2 9B	0.554	**0.575**	0.554	0.221	−0.292
Qwen3 8B Base	0.729	0.736	**0.743**	0.607	0.052
Qwen3.8 27B	0.600	**0.644**	0.402	0.454	−0.196
Pythia 6.9B†	0.479	**0.501**	−0.343	0.217	0.096

Bold marks the largest point estimate within each model, not a significant difference. †Pythia's attribute rate excludes 413 test rows with no predictions surviving the attribute filter; the other models have no undefined rates.

Readout definitions. For subject state s_e, reference states s_i, and reference Jacobians J_i, the differential and raw Jacobian readouts decode $\mathrm{mean}_i\, J_i(s_e - s_i)$ and $\mathrm{mean}_i\, J_i s_e$, respectively. Logit lens decodes s_e; differential logit lens decodes $s_e - \mathrm{mean}_i\, s_i$. Same-subject references are excluded, and final normalization is applied once before the vocabulary head. Each relation's score is the negative median $\log_{10}(\mathrm{rank} + 1)$ of the model's answer token.

Attribute rate. Following Geva et al. (2023) and the reconstruction of Gerstner et al. (2026), we measure overlap between filtered top-50 logit-lens predictions and subject attributes retrieved from Wikipedia. The index uses the union of current test subjects; the source stopword, length, and retrieval filters are retained. Relation scores average the percentage overlap. Undefined rates are handled as noted in the table; matched-row sensitivity results remain in the accompanying data.

Raw Jacobian has a larger point estimate than differential Jacobian in five models. All seven paired 95% relation-bootstrap intervals for their difference include zero; these results do not establish a general advantage for either readout. LRE hyperparameter selection remains in sample.

A.1 OLMo CORPUS-FREQUENCY BASELINES

Table 3 uses the completed scan of OLMo's three published training mixes: 175,877 shards and 3.86 billion document records. The counts are reused, while all aggregates and correlations are recomputed on the current shared-split test rows. They describe the published document multiset, rather than exact packed training sequences or training exposure weights.

Table 3: **Frequency baselines for OLMo 3 7B.** Correlations with LRE faithfulness across the same 25 shared-split relations, using document counts from OLMo's three published training mixes combined.

Frequency measure	Spearman ρ	p-value
Marginal frequencies		
Subject frequency, c_s	0.290	0.160
Object frequency, c_o	0.644	5.08×10^{-4}
Co-occurrence		
Joint frequency, c_{so}	0.578	0.002
Mean of $\log(1 + c_{so})$	0.568	0.003
$\log(1 + \text{mean}(c_{so}))$	0.578	0.002
Conditional frequencies		
Forward conditional, $P(o \mid s)$	0.386	0.056
Reverse conditional, $P(s \mid o)$	−0.328	0.109
Directional difference, $P(o \mid s) - P(s \mid o)$	0.362	0.075
Micro-averaged $P(o \mid s)$	0.615	0.001
Micro-averaged $P(s \mid o)$	0.030	0.887

Counts denote binary document presence. Means use the current test rows with both marginals observed; micro averages are ratios of summed counts. The p-values are two-sided and unadjusted.

Counting and aggregation. A document contributes at most one mention per subject/object string. Matching uses normalized literal names with Unicode word boundaries, without alias resolution or a proximity constraint. With subject, object, and joint counts c_s, c_o, and c_{so}, the conditionals are $P(o \mid s) = c_{so}/c_s$ and $P(s \mid o) = c_{so}/c_o$. Macro features average row-level quantities where both marginals are observed. Micro conditionals divide summed joint counts by summed marginals. The log of the mean joint count has the same Spearman correlation as the mean joint count because the transformation is monotonic.

Interpretation and reproducibility. These are exploratory associations. Relations can share subjects and objects, and the reported p-values are unadjusted. Conditional direction also reflects marginal frequency: when $c_{so} > 0$, $P(o \mid s)/P(s \mid o) = c_o/c_s$. The full measurements, Pearson results, coverage sensitivities, paired intervals, and source hashes remain in `code/results/shared_baselines/`. The tables are generated directly from the verified correlation records.

A.2 NEUTRAL-PROMPT VARIATIONS

Table 4 collects the prompt experiments for the differential Jacobian lens. For Qwen3 8B Base, we compare the original prompt with six alternatives and the subsequently measured newline control. The checkpoint, layer 19, reference and test rows, relation-prompt answer tokens, and LRE outcomes are fixed within each protocol. The legacy split contains 28 relations; the shared split contains 25. The lower panel reports the original-to-newline comparison for all seven models on the shared split, using each model's fixed layer.

Table 4: **Neutral-prompt variations.** Spearman ρ between differential-Jacobian scores and retained LRE faithfulness. Panel A compares exact templates for Qwen3 8B Base; panel B isolates the newline-prefix comparison across all seven models on the shared split.

A. Qwen3 8B Base: prompt wording and prefix

Neutral prompt	Shared ($n = 25$)	Legacy ($n = 28$)
`{s} is known for`	0.268	0.060
`{s}`	−0.057	0.058
`{s} is`	0.166	−0.118
`{s} can be described as`	0.372	0.179
`Information about {s}:`	0.733	0.295
`Facts about {s}:`	0.788	0.298
`Entity: {s} is known for`	0.728	0.202
`\n{s} is known for`	0.729	0.216

B. All models: newline prefix, shared split ($n = 25$)

Model	Original	Newline	$\Delta\rho$
OLMo 3 7B	0.624	0.683	+0.058
GPT-J 6B	0.155	0.573	+0.418
Mistral 7B	0.625	0.606	−0.018
Gemma 2 9B	0.487	0.554	+0.066
Qwen3 8B Base	0.268	0.729	+0.461
Qwen3.8 27B	0.596	0.600	+0.003
Pythia 6.9B	−0.045	0.479	+0.524

s denotes the subject; `\n` denotes one leading newline. Panel B compares `{s} is known for` with `\n{s} is known for`. The newline is a position control measured after the seven-template Qwen comparison; no additional newline is implied for the other templates.

Interpretation. Qwen's shared-split correlation rises from 0.268 under the original prompt to 0.729 with a leading newline; `Facts about {s}:` gives 0.788. These are exploratory sensitivity results, not held-out prompt selection. Prefix wording can change both context and subject position. The newline preserves Qwen's original content token IDs, while Mistral also retokenizes the initial content. The legacy and shared protocols are not independent replications, and no layer or LRE hyperparameter is retuned between prompts.

B JACOBIAN–VECTOR PRODUCTS

Neither the operator nor the lens ever needs J_i itself, only products $J_i u$: for the lens this is explicit in Eq. 4, and for the LRE it is what the expansion in Eq. 3 buys, since it turns $W = \mathrm{mean}_i\, J_i$ into a sum of such products. We obtain each one by the standard double-backward construction (Pearlmutter, 1994): differentiate $\langle u', f(h)\rangle$ with respect to h with u' held as a free variable, then differentiate the result with respect to u' in the direction u. Each product therefore costs a constant number of forward and backward passes rather than the d_{model} passes a materialized Jacobian would need. The construction requires `float32` and eager attention, as the fused attention kernels are not twice differentiable.

C LAYER SELECTION

Layer 17 is roughly half-way through the network, the region in which such lenses are usable at all (Gurnee et al., 2026), and is the modal best layer for the LRE itself in our sweep (10 of 28 relations; the rest spread over 5, 11, 14, 20 and 23). A single layer is used for *every* relation, without per-relation tuning. However, this does not establish robustness to the layer choice: changing the layer can affect relations differently. A systematic sensitivity analysis is left to future work.

D PERMUTATION TEST FOR THE MAIN CORRELATION

The correlation in §5 rests on 28 points, and its parametric p-value assumes more than we would like. We therefore also test it by permutation: we shuffle the lens scores against the faithfulness values 10^4 times, recompute ρ each time, and count how often the shuffled $|\rho|$ matches or exceeds the observed one. It does so in 0.03% of shuffles, so the association is not an artifact of the small sample.

E NUMERIC-OBJECT TOKENIZATION CORRECTION

Both relations with numeric objects (PRESIDENTS_BIRTH_YEAR, PRESIDENTS_ELECTION_YEAR) initially scored faithfulness 1.000. This was spurious. In this tokenizer `" 1743"` segments as `[' ', '174', '3']`: digits never merge with a preceding space. First-token scoring therefore compared the model's top-1 against a bare space token on both sides, and faithfulness 1.000 meant only that the LRE and the model both emit a space.

The fix is to place the space in the prompt, so that the query ends with it and the next token is content. Rescored, the two relations give faithfulness 0.67 and lens scores of -4.24 and -4.17; the model does predict the year correctly once the space is consumed. Including the uncorrected values would have depressed the reported correlation from 0.658 to 0.427.

F ALL RELATIONS

Table 5 gives faithfulness and the lens score for every relation in the study. Table 1 gives object prominence and the lens's top-five hit rate for the 14 relations on which prominence can be measured (Section 6).

Table 5: All 28 relations, sorted by LRE faithfulness. k is the mean number of subjects per distinct object; A_r is the relation-agnostic lens score, higher meaning closer agreement with the model. †rescored for the numeric-object artifact (Appendix E); *fewer than ten test items, so faithfulness is coarsely quantized.

relation	k	faith.	A_r
country_language*	1.7	1.00	-1.04
country_largest_city*	1.0	1.00	-0.81
landmark_on_continent	189.4	0.94	-2.81
person_native_language	30.6	0.94	-2.56
name_birthplace	3.9	0.93	-1.29
name_religion	6.2	0.93	-1.37
person_plays_pro_sport	63.6	0.92	-1.45
person_plays_instrument	85.5	0.88	-1.77
landmark_in_country	9.2	0.84	-2.25
city_in_country	1.3	0.82	-0.48
person_occupation	26.5	0.76	-2.81
country_capital_city*	1.0	0.75	-0.74
name_gender*	9.5	0.67	-4.67
presidents_birth_year†	1.3	0.67	-4.24
presidents_election_year†	1.1	0.67	-4.17
product_by_company	17.4	0.66	-3.24
person_plays_position_in_sport	68.0	0.64	-3.99
person_band_lead_singer*	1.0	0.60	-1.68
person_university	1.3	0.59	-4.52
company_hq	4.1	0.52	-4.09
superhero_person	1.0	0.48	-2.15
star_constellation	11.7	0.42	-3.67
pokemon_evolutions	1.0	0.39	-3.44
country_currency	1.3	0.36	-4.48
superhero_archnemesis	1.1	0.25	-3.57
person_father	1.0	0.09	-3.69
company_ceo	1.0	0.07	-4.59
person_mother	1.0	0.07	-3.99